Kaizhang Kang

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Research Interests

I am broadly interested in computational imaging. During my Ph.D. study, I focused on appearance/geometry acquisition & modeling. I developed a series of high-performance acquisition hardware systems from low-level circuit design to high-level control software. Based on proposed differentiable acquisition framework, the published works mainly adress how to efficiently and accurately digitize real objects.

As a postdoctoral researcher in Professor Seung-Hwan Baek's group, I have taken the opportunity to broaden my research into microscopy, spectral imaging, and other areas of computational imaging.

Looking ahead, I aim to explore novel and unconventional directions in computational imaging, pushing the limits of the field.

Education

Sep. 2018 - June 2023 **Zhejiang University**

Integrated Master and Ph.D. Program in Computer Science
(Supervised by Hongzhi Wu)

Sep. 2014 - June 2018 **Zhejiang University**

B.S. in Computer Science
Honors Degree from Chu Kochen Honors College

Work Experience

Apr. 2026 - Now Pohang University of Science and Technology
Postdoc

My main research project is about designing and building a high-speed Fourier ptychographic microscopy system and other computational imaging projects.

Feb. 2024 - Feb. 2026 King Abdullah University of Science and Technology
Postdoc

My main research project is about designing and building a lightweight spectral camera for determining coral health status. I also work on high-speed Fourier ptychographic microscopy system and other computational imaging projects.

Aug. 2022 - Jan. 2023 Meta Reality Labs
Research Scientist Intern.

The project is to estimate appearance of human head with multi-view images under any lighting conditions.

Publications

2026

- **CoralSpec-30M: A Large-Scale Coral Spectral Dataset Covering Multiple Species, Scales, and Health Conditions**
  Kaizhang Kang, *Wolfgang Heidrich*
  Limnology and Oceanography Letters† *, 11, 4 (July 2026), e70156.*
  † Q1 in Oceanography

- **Snapshot Polarimetric Display Inverse Rendering**

  *Seokjun Choi, Yunseong Moon, **Kaizhang Kang**, Hoon-Gyu Chung, Jin-Nyeong Kim, Giljoo Nam, Seung-Hwan Baek*
  *Accepted to **SIGGRAPH Asia 2026** (Journal)*

- **Low Latency Gaze Tracking via Latent Optical Sensing**

  *Yidan Zheng, Matheus Souza, **Kaizhang Kang**, Qiang Fu, Hadi Amata, Wolfgang Heidrich*
  *Accepted to **SIGGRAPH Asia 2026** (Conference)*

- **LoBoFit: Flexible Garment Refitting via Local Bone Mapping Blending**

  *Meng Zhang, Yu Xin, Feiya Guo, **Kaizhang Kang**, Mengyu Chu, Ruizhen Hu*
  *Proc. **SIGGRAPH 2026***

2025

- **High-throughput space-time Fourier ptychography for motile microorganisms**

  Ming Sun, **Kaizhang Kang***, Yogeshwar Mishra, Xinge Yang, Hadi Amata, Wolfgang Heidrich*
  ***Optics Express**† Vol. 33, Issue 19, pp. 39438-39451*
  * Contributed equally
  † Q1 in Optics

- **High-Speed Fourier Ptychographic Microscopy System for Live Microorganisms Imaging**

  ***Kaizhang Kang**, Ming Sun, Yogeshwar Mishra, Wolfgang Heidrich*
  *Proc. **Optica Imaging Congress 2025 (COSI)***

- **Designing and Fabricating Color BRDFs with Differentiable Wave Optics**

  *Yixin Zeng, Kiseok Choi, Hadi Amata, **Kaizhang Kang**, Wolfgang Heidrich, Hongzhi Wu, Min H. Kim*
  *ACM Trans. Graph. (Proc. **SIGGRAPH Asia 2025**), 44, 6 (Dec. 2025), 241.*

- **Efficient Depth- and Spatially-Varying Image Simulation for Defocus Deblur**

  *Xinge Yang, Chuong Nguyen, Wenbin Wang, **Kaizhang Kang**, Wolfgang Heidrich, Xiaoxing Li*
  *Proc. **ICCV Workshops** (ICCVW), 2025, pp. 3047-3057*

- **Latent Space Imaging**

  *Matheus Souza, Yidan Zheng, **Kaizhang Kang**, Yogeshwar Nath Mishra, Qiang Fu, Wolfgang Heidrich*
  *Proc. **CVPR 2025***

- **Learning Photometric Feature Transform for Free-form Object Scan**

  *Xiang Feng, **Kaizhang Kang**, Fan Pei, Huakeng Ding, Jinjiang You, Ping Tan, Kun Zhou and Hongzhi Wu*
  *IEEE **TVCG**, 31, 9 (Sep. 2025), pp. 6398-6409*

2023

- **Differentiable Dynamic Visible-Light Tomography**

  ***Kaizhang Kang**, Zoubin Bi, Xiang Feng, Yican Dong, Kun Zhou and Hongzhi Wu*
  *Proc. **SIGGRAPH Asia 2023***

- **Neural Reflectance Capture in the View-Illumination Domain**

  ***Kaizhang Kang**, Minyi Gu, Cihui Xie, Xuanda Yang, Hongzhi Wu and Kun Zhou*
  *IEEE **TVCG**, 29, 2 (Feb. 2023), pp. 1450-1462*

2021

- **Learning Efficient Photometric Feature Transform for Multi-view Stereo**

  ***Kaizhang Kang**, Cihui Xie, Ruisheng Zhu, Xiaohe Ma, Ping Tan, Hongzhi Wu and Kun Zhou*
  ICCV 2021

- **Free-form Scanning of Non-planar Appearance with Neural Trace Photography**

  *Xiaohe Ma, **Kaizhang Kang**, Ruisheng Zhu, Hongzhi Wu and Kun Zhou*
  *ACM Trans. Graph. (Proc. **SIGGRAPH 2021**), 40, 4 (Aug. 2021), 124.*

2019

- **Learning Efficient Illumination Multiplexing for Joint Capture of Reflectance and Shape**

  ***Kaizhang Kang**, Cihui Xie, Chengan He, Mingqi Yi, Minyi Gu, Zimin Chen, Kun Zhou and Hongzhi Wu*
  *ACM Trans. Graph. (Proc. **SIGGRAPH Asia 2019**), 38, 6 (Nov. 2019), 165.*

2018

- **Efficient Reflectance Capture Using an Autoencoder**

  ***Kaizhang Kang**, Zimin Chen, Jiaping Wang, Kun Zhou and Hongzhi Wu*
  *ACM Trans. on Graphics (Proc. **SIGGRAPH 2018**), 37, 4 (Aug. 2018), 127.*

Honors & Awards

Microsoft Research Asia Fellowship 2021

ACM SIGGRAPH Student Research Competition (2nd Place, Undergraduate Category) 2018

Lu Zengyong CAD&CG High Technology Award (2nd Place) 2019

Skills

- **Computer vision & graphics.** My research in the Ph.D. study mainly focuses on computer vision & graphics about how to digitize 3D objects in both high efficiency and high quality manner. I am also familiar with multi-view stereo, photometric stereo, reflectance modeling, and rendering algorithms including ray tracing and rasterization.
- **Hardware design.** I built multiple hardware prototypes from scratch, including circuit design, FPGA programming and mechanical design.

Invited Talks

Nov. 2025 Computer Graphics Lab., Pohang University of Science and Technology
Toward Intelligent Imaging Systems: From Appearance Acquisition to Multi-scale and Multi-modal Imaging

Feb. 2023 Shanghai Artificial Intelligence Laboratory
Differentiable Acquisition of Appearance & Shape

Nov. 2022 Computer Graphics Group (Julie Dorsey & Holly Rushmeier Lab), Yale University
Differentiable Acquisition of Appearance & Shape

Mar. 2022 Smart Geometry Processing Group (Niloy Mitra Lab), University College London (UCL)
Differentiable Acquisition of Appearance & Shape

Dec. 2019 Graphics and Mixed Environment Seminar (GAMES, Online)
Learning Efficient Illumination Multiplexing for Joint Capture of Reflectance and Shape

Academic Service

Committee Membership SIGGRAPH Asia Technical Communications & Posters Committee (2024, 2025)

Reviewer SIGGRAPH, SIGGRAPH Asia, ICCV, WACV, AAAI, etc.